

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
H H T H H H T H H T
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.
5. Given:
Input (x) = 4
Weight (w) = 2
Actual Output = 10
Prediction:
 $y = w \times x$
 1. Find the predicted output.
 2. Calculate the error.
 3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

*I **S.Mahammed Nayeem** (192425294) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

S.Mahammed Nayeem

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1: Coin Toss

Given:

- Total number of coin tosses (n) = 10
- Number of Heads (H) = 7

Formula:

$$\hat{p} = \text{Number of Heads} / \text{Total Tosses}$$

Calculation:

$$\hat{p} = 7 / 10 = 0.7$$

Answer:

The Maximum Likelihood Estimate (MLE) of the probability of getting Heads is **0.7**.

Question 2: Students Passing an Exam

Given:

- Total number of students = 50
- Number of students passed = 42

Formula:

$$\hat{p} = \text{Number of Passes} / \text{Total Students}$$

Calculation:

$$\hat{p} = 42 / 50 = 0.84$$

Answer:

The Maximum Likelihood Estimate (MLE) of the probability that a student passes is **0.84**.

Question 3: Defective Light Bulbs

Given:

- Total number of bulbs = 100
- Number of defective bulbs = 8

Formula:

$$\hat{p} = \text{Number of Defective Bulbs} / \text{Total Bulbs}$$

Calculation:

$$\hat{p} = 8 / 100 = 0.08$$

Answer:

The Maximum Likelihood Estimate (MLE) of the probability that a bulb is defective is **0.08**.

Question 4: Biased Die

Given:

- Total number of rolls = 60
- Number of times 6 appeared = 18

Formula:

$$\hat{p} = \text{Number of Successful Outcomes} / \text{Total Trials}$$

Calculation:

$$\hat{p} = 18 / 60 = 0.30$$

Answer:

The Maximum Likelihood Estimate (MLE) of the probability of rolling a 6 is **0.30**.

Question 5: Neural Network Weight Update

Given:

- Input (x) = 4
- Weight (w) = 2
- Actual Output = 10
- Gradient = 4
- Learning Rate (η) = 0.01

Formula 1: Predicted Output

$$y = w \times x$$

Calculation:

$$y = 2 \times 4 = 8$$

Answer:

Predicted Output = **8**

Formula 2: Error

Error = Actual Output – Predicted Output

Calculation:

$$\text{Error} = 10 - 8 = 2$$

Answer:

$$\text{Error} = 2$$

Formula 3: Updated Weight

$$\text{New Weight} = \text{Old Weight} - (\text{Learning Rate} \times \text{Gradient})$$

Calculation:

$$\text{New Weight} = 2 - (0.01 \times 4)$$

$$\text{New Weight} = 2 - 0.04 = 1.96$$

Answer:

The updated weight after one gradient descent step is **1.96**.